Parametric Equations: Cram Sheet

PCBC H | quiz 10/15

Posted: 4 questions, 30 minutes total, 12 points, 3 without a calculator and 1 with. The four types below are a **practice framework** built from the notes and worksheets; the real question types and order were not posted.

The four types

1 Eliminate t, restrict, graph

1. Solve the simpler equation for t.
2. Substitute; simplify.
3. t-table: ends + any turning point.
4. State the x and y values **actually reached**.
5. Graph with dots and direction arrows.

2 Trig curve

1. Isolate: $\cos t = \frac{x-h}{a}$, $\sin t = \frac{y-k}{b}$ (or $\tan, \sec$ / $\cot, \csc$).
2. Square and use the identity.
3. Pick the branch: $\sec t > 0$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, $\csc t > 0$ on $(0, \pi)$.
4. t-table for direction.

3 Two parametrizations

1. $x = t$, $y = f(t)$.
2. Complete the square: $y = (x-h)^2 + k$.
3. $t = x - h$, so $x = t + h$, $y = t^2 + k$.
4. Write "for t in $(-\infty, \infty)$" with both (when every x is allowed).

4 Projectile (calculator, degree mode)

$x = (v\cos\theta)t, \quad y = -16t^2 + (v\sin\theta)t + h$

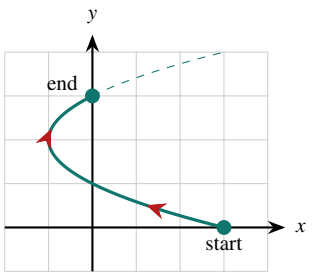
1. Flight time: $y = 0$, quadratic formula, reject negative t.
2. Distance: **unrounded** t into x.
3. Top: $t = \frac{v\sin\theta}{32}$, then y.
4. Rectangular: $t = \frac{x}{v\cos\theta}$ $(v\cos\theta \neq 0)$.

Identities

you see	use	you get
$\sin t$, $\cos t$	$\sin^2 t + \cos^2 t = 1$	circle or ellipse
$\tan t$, $\sec t$	$\sec^2 t - \tan^2 t = 1$	hyperbola, upper branch on $(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot t$, $\csc t$	$\csc^2 t - \cot^2 t = 1$	hyperbola, upper branch on $(0, \pi)$
$\sin t$, $\csc t$	$\csc t = \frac{1}{\sin t}$	$y = \frac{k}{x}$ piece

Arrows, ends and restrictions

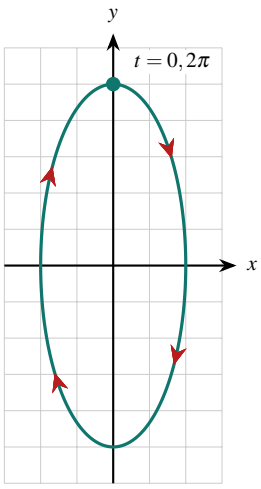
- **Red arrows along the curve**: the way t increases. Check with a t-table: start, middle, end.
- **Arrowhead at the very end**: the curve keeps going (x or $y \to \pm\infty$).
- **Solid dot**: endpoint reached (closed bracket).
- **Open circle**: excluded endpoint, *only if* no other allowed t reaches that point.
- **Closed curve**: an ellipse over $[0, 2\pi]$ starts and ends at one point.
- $\sqrt{}$ gives $x \geq 0$. t^2, $\sin t$, $\cos t$ can turn around inside the interval: check the middle. Carry $t \neq c$ over to x.



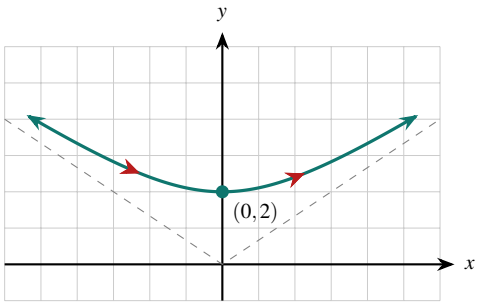
$x = t^2 - 1$, $y = t + 2$, $[-2, 1]$
$-1 \leq x \leq 3$, $0 \leq y \leq 3$

Curve shapes and direction

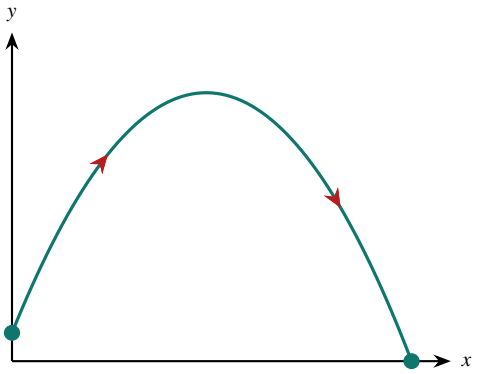
sideways parabola	$x = (y-k)^2 + h$, from $x = t^2$ and y linear in t
parabola up/down	$y = a(x-h)^2 + k$, from x linear in t and $y = t^2$; projectiles open down
circle, ellipse	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$. With $a, b > 0$: $a\cos t,\ b\sin t$ counterclockwise from $(a,0)$; $a\sin t,\ b\cos t$ clockwise from $(0,b)$. One sign flip reverses it; two keep it.
hyperbola branch	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$, keep $y = +\sqrt{\cdot}$. tan / sec: left to right. cot / csc: right to left. Asymptotes $y = \pm\frac{a}{b}x$.
reciprocal	$y = \frac{k}{x}$; $x = 2\sin t$, $y = 4\csc t$ on $(0,\pi)$ retraces: down to $(2,4)$, back up.



$x = 2\sin t,\ y = 5\cos t$
clockwise, one start/end



$x = 3\tan t,\ y = 2\sec t$
upper branch $y \geq 2$, left to right



$v = 128$, $24°$, $h = 5$: lands $t \approx 3.347$,
$x \approx 391.407$ ft

Red arrows: direction as t increases. Plain arrowheads at the ends of the hyperbola: the curve keeps going.

Traps

- Only plugging the t endpoints into x when x turns around inside the interval.
- Both hyperbola branches, or $y = \pm\sqrt{\cdot}$, when $\sec t$ or $\csc t$ is always positive.
- Leaving off the t interval on a parametrization; $t = x + 3$ means $x = t - 3$.
- Rounding t before finding x; radian mode with degree angles.

Sources: combined parametric notes, day 1 and day 2 worksheets, warm-up after day 1, the Before Parametrics page (conics). Separate day 2 notes and the warm-up before parametrics (PA) were not available.